

TECNOLOGICO NACIONAL DE MEXICO

INSTITUTO TECNOLOGICO DE PUEBLA

Ingeniería en Tics

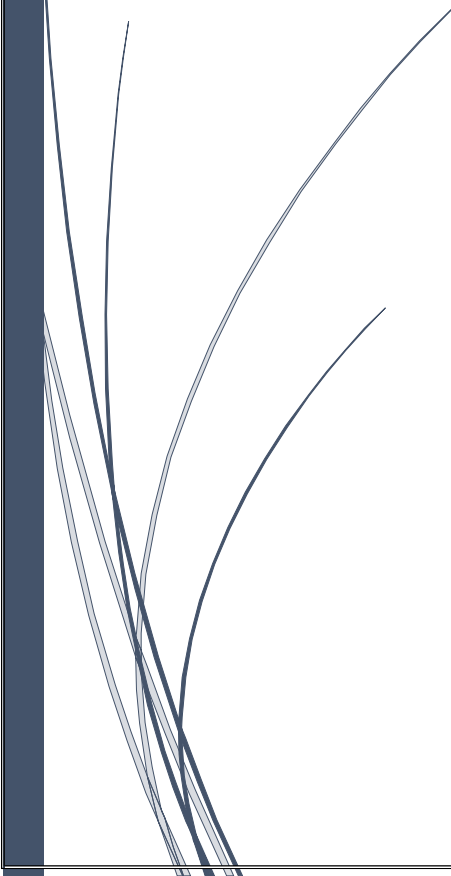
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Entrega: 24/05/26



Introducción

Este proyecto es la gestión hospitalaria moderna, la disponibilidad, consistencia y segregación de los datos representan pilares críticos para garantizar tanto la eficiencia operativa como la seguridad en la atención de los pacientes. El presente proyecto aborda el diseño, la implementación y la automatización de un sistema de información hospitalario distribuido y heterogéneo, estructurado mediante una arquitectura de microservicios de datos basada en contenedores Docker.

Docker-compose.yml

```
version: '3.8'

services:
  mysql_urgencias:
    image: mysql:8.0
    container_name: mysql_urgencias
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root_password
      MYSQL_DATABASE: urgencias_db
    ports:
      - "3306:3306"
    volumes:
      - mysql_data:/var/lib/mysql

  sqlserver_farmacia:
    image: mcr.microsoft.com/mssql/server:2022-latest
    container_name: sqlserver_farmacia
    restart: always
    environment:
      ACCEPT_EULA: "Y"
      MSSQL_SA_PASSWORD: "Your_Password123"
    ports:
      - "1433:1433"
    volumes:
      - mssql_data:/var/opt/mssql

  postgres_finanzas:
    image: postgres:15
    container_name: postgres_finanzas
    restart: always
    environment:
      POSTGRES_USER: postgres
      POSTGRES_PASSWORD: postgres_password
      POSTGRES_DB: finanzas_auditoria_db
    ports:
      - "5432:5432"
    volumes:
      - postgres_data:/var/lib/postgresql/data
```

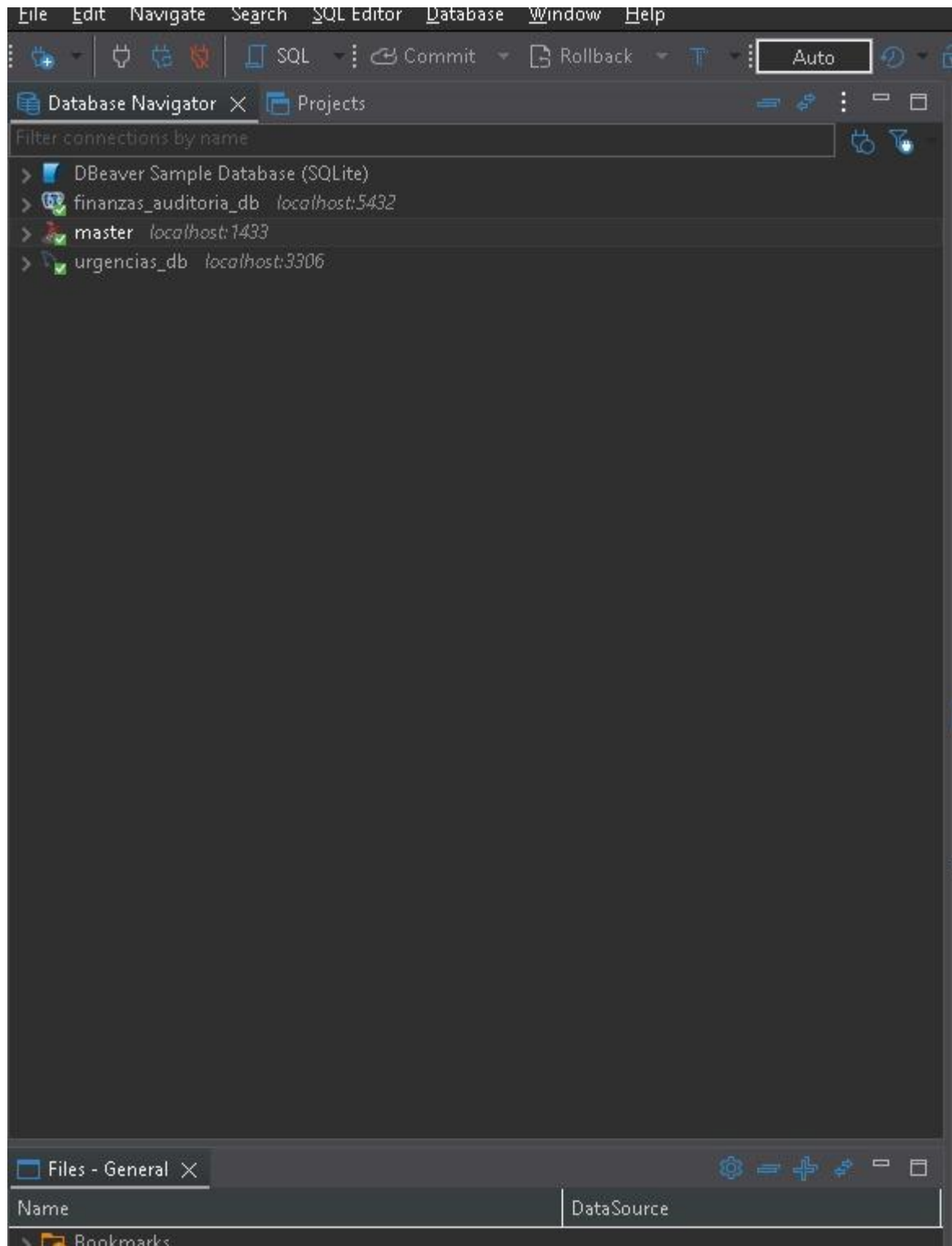
Conectividad del Docker Compose

The screenshot displays the Docker Desktop interface for a project named "proyectorico". The interface is divided into several sections:

- Top Bar:** Shows the project name "proyectorico" and the path "C:\Users\HP\Desktop\Proyecto Nico". It includes a "View configurations" button and icons for play, stop, and delete.
- Containers List:** On the left, a list of containers is shown with their names and ports:
 - sqlserver_far...** (mysql/server:2022) with port 1433:1433.
 - postgres_fina...** (postgres:15) with port 5432:5432.
 - mysql_urgenc...** (mysql:8.0) with port 3306:3306.
 - sqlserver_farnacta**
 - mysql_urgencias**
 - postgres_finanzas**
- Logs View:** The main area displays logs for the selected container, "mysql_urgencias". The logs show the following sequence of events:
 - 2026-05-24 21:25:20 UTC [1] LOG: database system is shut down
 - SQL Server 2022 will run as non-root by default. This container is running as user mssql. Your master database file is owned by mssql. To learn more visit <https://go.microsoft.com/fwlink/?linkid=2099216>.
 - 2026-05-24 21:34:29+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
 - PostgreSQL Database directory appears to contain a database, Skipping initialization
 - 2026-05-24 21:34:30.514 UTC [1] LOG: starting PostgreSQL 15.10 (Debian 15.10-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
 - 2026-05-24 21:34:30.517 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
 - 2026-05-24 21:34:30.517 UTC [1] LOG: listening on IPv6 address ":::", port 5432
 - 2026-05-24 21:34:30.553 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
 - 2026-05-24 21:34:30.596 UTC [30] LOG: database system was shut down at 2026-05-24 21:25:20 UTC
 - 2026-05-24 21:34:30.664 UTC [1] LOG: database system is ready to accept connections
 - 2026-05-24 21:34:31+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
 - 2026-05-24 21:34:31+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
 - 2026-05-24 21:34:32.297517Z [0] [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
 - 2026-05-24 21:34:32.303746Z [0] [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
 - 2026-05-24 21:34:32.341006Z [1] [System] [MY-013576] [InnoDB] InnoDB initialization has started.
 - 2026-05-24 21:34:33.972113Z [1] [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
 - 2026-05-24 21:34:34.827549Z [0] [System] [MY-010229] [Server] Starting XA crash recovery...
 - 2026-05-24 21:34:34.883987Z [0] [System] [MY-010232] [Server] XA crash recovery finished.
 - 2026-05-24 21:34:35.104564Z [0] [Warning] [MY-010968] [Server] CA certificate ca.pem is self signed.
 - 2026-05-24 21:34:35.104604Z [0] [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
 - 2026-05-24 21:34:35.149708Z [0] [Warning] [MY-011010] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
 - 2026-05-24 21:34:35.296021Z [0] [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: ':::' port: 33060, socket: /var/run/mysqld/mysqld.sock
 - 2026-05-24 21:34:35.297306Z [0] [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.

At the bottom, system resources are shown: RAM 2.72 GB, CPU 49.87%, Disk 5.64 GB used (limit 1006.95 GB). The terminal window at the bottom right shows the command prompt and the version "v4.74.0". The system clock indicates 03:34 p.m. on 24/05/2026.

Evidencias de DBaver Conexión



Creación De Tablas

```
CREATE TABLE pacientes (
  nss VARCHAR(15) PRIMARY KEY,
  nombre VARCHAR(100) NOT NULL,
  fecha_nacimiento DATE NOT NULL,
  tipo_sangre VARCHAR(5) NOT NULL
);

CREATE TABLE especialidades (
  id_especialidad INT AUTO_INCREMENT PRIMARY KEY,
  nombre VARCHAR(100) NOT NULL,
  piso INT NOT NULL
);

CREATE TABLE medicos (
  cedula VARCHAR(20) PRIMARY KEY,
  nombre VARCHAR(100) NOT NULL,
  id_especialidad INT NOT NULL,
  turno VARCHAR(20) NOT NULL,
  FOREIGN KEY (id_especialidad)
    REFERENCES especialidades(id_especialidad)
);

CREATE TABLE citas (
  id_cita INT AUTO_INCREMENT PRIMARY KEY,
  nss_paciente VARCHAR(15) NOT NULL,
  cedula_medico VARCHAR(20) NOT NULL,
  fecha_hora DATETIME NOT NULL,
  estado VARCHAR(20) NOT NULL,
  FOREIGN KEY (nss_paciente)
    REFERENCES pacientes(nss),
  FOREIGN KEY (cedula_medico)
    REFERENCES medicos(cedula)
);

CREATE TABLE triaje_urgencias (
  id_triaje INT AUTO_INCREMENT PRIMARY KEY,
```

```
CREATE TABLE seguros (
  id_seguro SERIAL PRIMARY KEY,
  aseguradora VARCHAR(150) NOT NULL,
  cobertura_maxima DECIMAL(12, 2) NOT NULL
);

CREATE TABLE facturacion (
  id_factura SERIAL PRIMARY KEY,
  nss_paciente VARCHAR(15) NOT NULL,
  monto_total DECIMAL(12, 2) NOT NULL,
  id_seguro INT,
  fecha_emision TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_facturacion_seguros FOREIGN KEY (id_seguro) REFERENCES seguros(id_seguro)
);

CREATE TABLE historial_clinico (
  id_historial SERIAL PRIMARY KEY,
  nss_paciente VARCHAR(15) NOT NULL,
  diagnostico TEXT NOT NULL,
  estado_paciente VARCHAR(50) NOT NULL,
  fecha_registro TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE pagos_nomina (
  id_nomina SERIAL PRIMARY KEY,
  cedula_medico VARCHAR(20) NOT NULL,
  monto_pago DECIMAL(12, 2) NOT NULL,
  mes_anio VARCHAR(7) NOT NULL,
  fecha_transferencia TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE log_auditoria (
  id_log SERIAL PRIMARY KEY,
  modulo VARCHAR(100),
  accion VARCHAR(50),
```

```
CREATE TABLE categorias_med (
  id_categoria INT IDENTITY(1,1) PRIMARY KEY,
  nombre VARCHAR(100) NOT NULL
);

CREATE TABLE laboratorios (
  id_laboratorio INT IDENTITY(1,1) PRIMARY KEY,
  razon_social VARCHAR(150) NOT NULL,
  telefono VARCHAR(20)
);

CREATE TABLE medicamentos (
  id_medimento INT IDENTITY(1,1) PRIMARY KEY,
  nombre VARCHAR(150) NOT NULL,
  id_categoria INT NOT NULL,
  id_laboratorio INT NOT NULL,
  stock INT NOT NULL DEFAULT 0,
  precio DECIMAL(10, 2) NOT NULL,
  CONSTRAINT FK_medicamentos_categorias FOREIGN KEY (id_categoria)
    REFERENCES categorias_med(id_categoria),
  CONSTRAINT FK_medicamentos_laboratorios FOREIGN KEY (id_laboratorio)
    REFERENCES laboratorios(id_laboratorio)
);

CREATE TABLE recetas_surtidas (
  id_receta INT IDENTITY(1,1) PRIMARY KEY,
  nss_paciente VARCHAR(150) NOT NULL,
  id_medimento INT NOT NULL,
  cantidad INT NOT NULL,
  fecha_surtido DATETIME DEFAULT GETDATE(),
  CONSTRAINT FK_recetas_medicamentos FOREIGN KEY (id_medimento) REFERENCES medicamentos
);

CREATE TABLE equipamiento_medico (
  id_equipo INT PRIMARY KEY IDENTITY(1,1),
  nombre_equipo VARCHAR(100),
  area_asignada VARCHAR(100),
```

```

from faker import Faker
import random
from datetime import datetime, timedelta

```

```

import mysql.connector
import pyodbc
import psycopg2
from psycopg2.extras import Json

fake = Faker('es_MX')

```

```

mysql_conn = mysql.connector.connect(
    host="localhost",
    port=3306,
    user="root",
    password="root_password",
    database="urgencias_db"
)

```

```

mysql_cursor = mysql_conn.cursor()

```

```

sqlserver_conn = pyodbc.connect(
    "DRIVER={ODBC Driver 18 for SQL Server};"
    "SERVER=localhost,1433;"
    "DATABASE=farmacia_db;"
    "UID=sa;"
    "PWD=Your_Password123;"
    "TrustServerCertificate=yes;"
)

```

```

sqlserver_cursor = sqlserver_conn.cursor()

```

```

postgres_conn = psycopg2.connect(
    host="localhost",
    port=5432,
    user="postgres",
    password="postgres_password"
)

```

```

40     password="postgres_password",
41     database="finanzas_auditoria_db"
42 )
43
44 postgres_cursor = postgres_conn.cursor()
45
46 print("✅ Conectado a las 3 bases de datos")
47
48
49
50 def fecha_random(inicio=2022, fin=2026):
51     fecha_inicio = datetime(inicio, 1, 1)
52     fecha_fin = datetime(fin, 12, 31)
53
54     return fecha_inicio + timedelta(
55         seconds=random.randint(
56             0,
57             int((fecha_fin - fecha_inicio).total_seconds())
58         )
59     )
60

```

```

61 print("\nInsertando datos en MySQL...")
62
63
64

```

```

65 especialidades = [
66     ("Cardiologia", 2),
67     ("Pediatria", 3),
68     ("Neurologia", 4),
69     ("Traumatologia", 5),
70     ("Urgencias", 1),
71     ("Dermatologia", 6),
72     ("Oncologia", 7),
73     ("Psiquiatria", 8)
74 ]
75

```

```

76 for nombre, piso in especialidades:
77     mysql_cursor.execute("""
78         INSERT INTO especialidades(nombre, piso)

```

```

78         INSERT INTO especialidades(nombre, piso)
79         VALUES(%s, %s)
80     """, (nombre, piso))
81
82     mysql_conn.commit()
83
84     mysql_cursor.execute("SELECT id_especialidad FROM especialidades")
85     ids_especialidades = [x[0] for x in mysql_cursor.fetchall()]
86
87     tipos_sangre = ["A+", "A-", "B+", "B-", "AB+", "AB-", "O+", "O-"]
88
89     nss_pacientes = []
90
91     for _ in range(80):
92
93         nss = str(random.randint(1000000000, 9999999999))
94
95         nss_pacientes.append(nss)
96
97         mysql_cursor.execute("""
98             INSERT INTO pacientes(
99                 nss,
100                 nombre,
101                 fecha_nacimiento,
102                 tipo_sangre
103             )
104             VALUES(%s, %s, %s, %s)
105         """, (
106             nss,
107             fake.name(),
108             fake.date_of_birth(minimum_age=1, maximum_age=90),
109             random.choice(tipos_sangre)
110         ))
111
112     mysql_conn.commit()
113
114     cedulas_medicos = []
115
116     turnos = ["Matutino", "Vespertino", "Nocturno"]

```

```

for _ in range(20):

    cedula = str(random.randint(1000000, 9999999))

    cedulas_medicos.append(cedula)

    mysql_cursor.execute("""
        INSERT INTO medicos(
            cedula,
            nombre,
            id_especialidad,
            turno
        )
        VALUES(%s, %s, %s, %s)
    """, (
        cedula,
        fake.name(),
        random.choice(ids_especialidades),
        random.choice(turnos)
    ))

```

```

mysql_conn.commit()

```

```

estados_cita = [
    "Pendiente",
    "Completada",
    "Cancelada"
]

```

```

for _ in range(50):

    mysql_cursor.execute("""
        INSERT INTO citas(
            nss_paciente,
            cedula_medico,
            fecha_hora,
            estado
        )

```

```

        estado
    )
    VALUES(%s, %s, %s, %s)
""" , (
    random.choice(nss_pacientes),
    random.choice(cedulas_medicos),
    fecha_random(),
    random.choice(estados_cita)
))

mysql_conn.commit()

niveles = ["Leve", "Moderado", "Grave", "Crítico"]

sintomas_lista = [
    "Dolor de cabeza",
    "Fiebre",
    "Fractura",
    "Dolor abdominal",
    "Tos",
    "Dificultad respiratoria",
    "Mareo",
    "Convulsiones"
]

for _ in range(42):

    mysql_cursor.execute("""
        INSERT INTO triaje_urgencias(
            nss_paciente,
            nivel_gravedad,
            sintomas,
            fecha_ingreso
        )
        VALUES(%s, %s, %s, %s)
    """, (
        random.choice(nss_pacientes),
        random.choice(niveles),
        random.choice(sintomas_lista),

```

Ln 22

```

        random.choice(sintomas_lista),
        fecha_random()
    ))

mysql_conn.commit()

print("✅ MySQL poblado")

print("\nInsertando datos en SQL Server...")

categorias = [
    "Analgésicos",
    "Antibióticos",
    "Antiinflamatorios",
    "Antidepresivos",
    "Vitaminas",
    "Antihistaminicos"
]

for categoria in categorias:

    sqlserver_cursor.execute("""
        INSERT INTO categorias_med(nombre)
        VALUES(?)
    """, categoria)

sqlserver_conn.commit()

sqlserver_cursor.execute("SELECT id_categoria FROM categorias_med")
ids_categoria = [x[0] for x in sqlserver_cursor.fetchall()]

for _ in range(14):

    sqlserver_cursor.execute("""
        INSERT INTO laboratorios(
            razon_social,
            telefono
        )
        VALUES(?, ?)

```

Ln

```

        VALUES(?, ?)
    """, (
        fake.company(),
        fake.phone_number()
    ))

sqlserver_conn.commit()

sqlserver_cursor.execute("SELECT id_laboratorio FROM laboratorios")
ids_laboratorios = [x[0] for x in sqlserver_cursor.fetchall()]

medicamentos = [
    "Paracetamol",
    "Ibuprofeno",
    "Amoxicilina",
    "Loratadina",
    "Omeprazol",
    "Diclofenaco",
    "Aspirina",
    "Metformina"
]

for _ in range(50):

    sqlserver_cursor.execute("""
        INSERT INTO medicamentos(
            nombre,
            id_categoria,
            id_laboratorio,
            stock,
            precio
        )
        VALUES(?, ?, ?, ?, ?)
    """, (
        random.choice(medicamentos),
        random.choice(ids_categoria),
        random.choice(ids_laboratorios),
        random.randint(10, 500),
        round(random.uniform(50, 3000), 2)

```



```

269     ))
270
271     sqlserver_conn.commit()
272
273     sqlserver_cursor.execute("SELECT id_medimento FROM medicamentos")
274     ids_medamentos = [x[0] for x in sqlserver_cursor.fetchall()]
275
276     for _ in range(50):
277
278         sqlserver_cursor.execute("""
279             INSERT INTO recetas_surtidas(
280                 nss_paciente,
281                 id_medimento,
282                 cantidad,
283                 fecha_surtido
284             )
285             VALUES(?, ?, ?, ?)
286         """, (
287             random.choice(nss_pacientes),
288             random.choice(ids_medamentos),
289             random.randint(1, 5),
290             fecha_random()
291         ))
292
293     sqlserver_conn.commit()
294
295     equipos = [
296         "Rayos X",
297         "Resonancia Magnética",
298         "Ultrasonido",
299         "Ventilador",
300         "Electrocardiograma",
301         "Desfibrilador"
302     ]
303
304     areas = [
305         "Urgencias",
306         "Radiología",
307

```

```

308         "Radiología",
309         "Quirófano",
310         "UCI",
311         "Pediatria"
312     ]
313
314     for _ in range(30):
315
316         sqlserver_cursor.execute("""
317             INSERT INTO equipamiento_medico(
318                 nombre_equipo,
319                 area_asignada,
320                 fecha_ultimo_mantenimiento
321             )
322             VALUES(?, ?, ?)
323         """, (
324             random.choice(equipos),
325             random.choice(areas),
326             fake.date_between(start_date='-2y', end_date='today')
327         ))
328
329     sqlserver_conn.commit()
330
331     print("✅ SQL Server poblado")
332
333     print("\nInsertando datos en PostgreSQL...")
334
335     aseguradoras = [
336         "GNP",
337         "AXA",
338         "MetLife",
339         "Qualitas",
340         "Mapfre",
341         "BBVA Seguros"
342     ]
343
344     for aseguradora in aseguradoras:

```

```

345     for aseguradora in aseguradoras:
346
347         postgres_cursor.execute("""
348             INSERT INTO seguros(
349                 aseguradora,
350                 cobertura_maxima
351             )
352             VALUES(%s, %s)
353         """, (
354             aseguradora,
355             round(random.uniform(50000, 1000000), 2)
356         ))
357
358     postgres_conn.commit()
359
360     postgres_cursor.execute("SELECT id_seguro FROM seguros")
361     ids_seguros = [x[0] for x in postgres_cursor.fetchall()]
362
363     for _ in range(70):
364
365         postgres_cursor.execute("""
366             INSERT INTO facturacion(
367                 nss_paciente,
368                 monto_total,
369                 id_seguro,
370                 fecha_emision
371             )
372             VALUES(%s, %s, %s, %s)
373         """, (
374             random.choice(nss_pacientes),
375             round(random.uniform(1000, 50000), 2),
376             random.choice(ids_seguros),
377             fecha_random()
378         ))
379
380     postgres_conn.commit()
381
382     diagnosticos = [

```

```

383     "Gripe",
384     "Fractura",
385     "Diabetes",
386     "Hipertensión",
387     "Migraña",
388     "Ansiedad"
389 ]
390
391 estados = [
392     "Estable",
393     "En tratamiento",
394     "Crítico",
395     "Recuperado"
396 ]
397
398 for _ in range(60):
399
400     postgres_cursor.execute("""
401         INSERT INTO historial_clinico(
402             nss_paciente,
403             diagnostico,
404             estado_paciente,
405             fecha_registro
406         )
407         VALUES(%s, %s, %s, %s)
408     """, (
409         random.choice(nss_pacientes),
410         random.choice(diagnosticos),
411         random.choice(estados),
412         fecha_random()
413     ))
414
415     postgres_conn.commit()
416
417 for _ in range(40):
418
419     postgres_cursor.execute("""
420         INSERT INTO pagos_nomina(
421             cedula_medico,

```

```

421             cedula_medico,
422             monto_paid,
423             mes_anio,
424             fecha_transferencia
425         )
426         VALUES(%s, %s, %s, %s)
427     """, (
428         random.choice(cedulas_medicos),
429         round(random.uniform(15000, 80000), 2),
430         datetime.now().strftime("%Y-%m"),
431         fecha_random()
432     ))
433     postgres_conn.commit()
434
435
436     modulos = [
437         "Pacientes",
438         "Farmacia",
439         "Facturación",
440         "Usuarios",
441         "Inventario"
442     ]
443
444     acciones = [
445         "INSERT",
446         "UPDATE",
447         "DELETE",
448         "LOGIN"
449     ]
450
451     for _ in range(74):
452
453         detalles = {
454             "usuario": fake.user_name(),
455             "ip": fake.ipv4(),
456             "descripcion": fake.sentence()
457         }
458
459         postgres_cursor.execute("""
460             INSERT INTO log_actividad(

```

```

detalles = {
    "usuario": fake.user_name(),
    "ip": fake.ipv4(),
    "descripcion": fake.sentence()
}

postgres_cursor.execute("""
    INSERT INTO log_auditoria(
        modulo,
        accion,
        fecha,
        detalles
    )
    VALUES(%s, %s, %s, %s)
""", (
    random.choice(modulos),
    random.choice(acciones),
    fecha_random(),
    Json(detalles)
))

postgres_conn.commit()

print("✅ PostgreSQL poblado")

mysql_cursor.close()
mysql_conn.close()

sqlserver_cursor.close()
sqlserver_conn.close()

postgres_cursor.close()
postgres_conn.close()

print("\n=====")
print("✅ DATOS INSERTADOS CORRECTAMENTE")
print("=====")

```

Insertación de Datos

```

C:\Users\HP\Desktop\Proyecto Nico>python Datos_db.py
✅ Conectado a las 3 bases de datos

Insertando datos en MySQL...
✅ MySQL poblado

Insertando datos en SQL Server...
✅ SQL Server poblado

Insertando datos en PostgreSQL...
✅ PostgreSQL poblado

=====
✅ DATOS INSERTADOS CORRECTAMENTE
=====

C:\Users\HP\Desktop\Proyecto Nico>

```

Ejemplo de consultas

```

-- 1.
SELECT m.nombre AS medico, e.nombre AS especialidad
FROM medicos m
INNER JOIN especialidades e ON m.id_especialidad = e.id_especialidad;

-- 2.
SELECT e.nombre AS especialidad, COUNT(m.cedula) AS total_medicos
FROM especialidades e
LEFT JOIN medicos m ON e.id_especialidad = m.id_especialidad
GROUP BY e.id_especialidad, e.nombre;

-- 3.
SELECT p.nombre AS paciente, t.nivel_gravedad, t.fecha_ingreso
FROM pacientes p
INNER JOIN triaje_urgencias t ON p.nss = t.nss_paciente;

-- 4.
SELECT p.nombre AS paciente, m.nombre AS medico, c.fecha_hora AS fecha_cita
FROM citas c
INNER JOIN pacientes p ON c.nss_paciente = p.nss
INNER JOIN medicos m ON c.cedula_medico = m.cedula;

-- 5.
SELECT p.nss, p.nombre AS paciente
FROM pacientes p
LEFT JOIN triaje_urgencias t ON p.nss = t.nss_paciente
WHERE t.nss_paciente IS NULL;

-- 6.
SELECT e.nombre AS especialidad, COUNT(c.id_cita) AS total_citas
FROM especialidades e
INNER JOIN medicos m ON e.id_especialidad = m.id_especialidad
INNER JOIN citas c ON m.cedula = c.cedula_medico
GROUP BY e.id_especialidad, e.nombre;

```

Tiggers

```

1  -- 1.
2
3
4  DELIMITER $$
5
6  CREATE TRIGGER trg_validar_fecha_nacimiento
7  BEFORE INSERT ON pacientes
8  FOR EACH ROW
9  BEGIN
10     IF NEW.fecha_nacimiento > CURDATE() THEN
11         SIGNAL SQLSTATE '45000'
12         SET MESSAGE_TEXT = 'Error: La fecha de nacimiento no puede ser futura.';
13     END IF;
14 END$$
15
16 DELIMITER ;
17
18 -- 2.
19
20
21 DELIMITER $$
22
23 CREATE TRIGGER trg_mayusculas_critico
24 BEFORE INSERT ON triaje_urgencias
25 FOR EACH ROW
26 BEGIN
27     IF NEW.nivel_gravedad = 'Critico' THEN
28         SET NEW.sintomas = UPPER(NEW.sintomas);
29     END IF;
30 END$$
31
32 DELIMITER ;
33

```

```

34
35
36 DELIMITER ;
37
38 -- 4.
39
40 DELIMITER $$
41
42 CREATE TRIGGER trg_validar_empalme_citas
43 BEFORE INSERT ON citas
44 FOR EACH ROW
45 BEGIN
46     IF EXISTS (
47         SELECT 1
48         FROM citas
49         WHERE cedula_medico = NEW.cedula_medico
50             AND fecha_hora = NEW.fecha_hora
51     ) THEN
52         SIGNAL SQLSTATE '45000'
53         SET MESSAGE_TEXT = 'Error: El médico ya tiene una cita en esa fecha y hora.';
54     END IF;
55 END$$
56
57 DELIMITER ;
58

```